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SHORT-PAPER

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Poster: Combine Topology and Traffic to Calibrate P2P Botnet Identification in Large-Scale Network

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ABSTRACT

Cyber attacks are increasingly becoming prevalent and influence the Internet ecosystem. Particularly, botnets crafted by adversaries cause significant damage to Internet infrastructure. In the last decade, the boom of P2P protocols broadened and amplified the attack surface, manifested as the built botnet usually containing over 100K nodes. However, the existing state-of-the-art graph-based method exhibits many misclassifications when existing legitimate P2P communication. In this paper, we propose an intelligent node retrieval process based on reinforcement learning to calibrate more misidentifications with as few retrievals as possible. The empirical evaluations that contain >100,000 communication nodes from the real-world IP backbone topology and involve 7 common P2P protocols show our proposal realizes superior outcomes for misidentification calibrations.

CCS CONCEPTS

• **Networks** → **Network security**; **Network monitoring**.

KEYWORDS

P2P botnet, network topology, RL, traffic analysis

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1 INTRODUCTION

With the Internet becoming ubiquitous, cyber attacks occur frequently [16]. Especially botnets which adversaries could control hosts to carry out malicious activities and bring damage to Internet infrastructure, e.g., distributed denial-of-service (DDoS) attacks, spamming, and so on [17, 20–22, 25, 28, 29, 31, 35]. Most famously, Mirai was used to launch vast volumetric DDoS [1], e.g., Dyn (a DNS provider) suffered a 1.2 Tbps attack. Recently, the new variant

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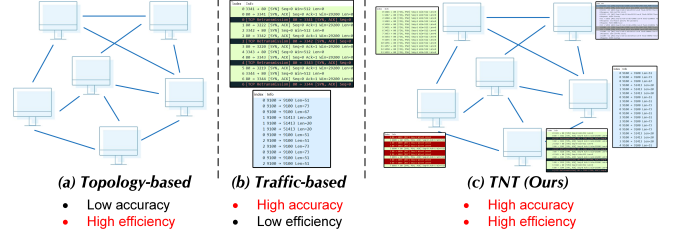


Figure 1: Illustrative explanation of combining topology and traffic analysis.

Hajime came into view of the community as it infected more than millions of devices based on the P2P protocol [11].

Currently, Graph Neural Networks (GNNs) are widely applied in large-scale botnet detection. They extract the communication topology and perform botnet node classification [19, 36]. However, existing work's assumptions [36] are somewhat limited given the lack of consideration that benign users may use P2P protocols to build legitimate businesses, e.g., BitTorrent is based on the Kademlia [18] Distributed Hash Table (DHT) protocol. Considering the more realistic scenario that exists both P2P-based botnets and legitimate P2P services, the SOTA model inevitably produce a lot of false positives and false negatives (i.e., low accuracy in Figure 1).

In this work, we study the real-world problem in which benign and malicious P2P protocol communication appears simultaneously. To calibrate misidentification for botnets, a vanilla design could investigate traffic [7, 23, 32, 33] for each host [3, 5, 8–10, 12, 30], but this is generally unrealistic due to the mass network nodes [34] (i.e., low efficiency in Figure 1). If the most likely misidentified nodes can be directly discovered based on graph knowledge, the calibration efficiency will be greatly improved. Therefore, we propose an intelligent node retrieval process based on reinforcement learning (RL) to tackle this issue. The desired result is to calibrate for more misidentifications with as few retrievals as possible.

Overall, the primary **contribution** of this paper is investigating the limitations of the SOTA graph-based botnet detection method when existing legitimate P2P communications, as well as designing an RL-based node retrieval process to efficiently calibrate error recognitions. Through extensive evaluation with topologies containing >100,000 nodes from real-world IP backbone, and involving 7 common P2P protocols, we demonstrate that the proposal has significant performance.

2 METHODOLOGY

We use Graph Convolutional Network (GCN) [15] to model the topology and leverage RL to select candidate hosts to analyze traffic. **Environment and State.** The environment refers to a series of information sets and the state as the input will be fed to the agent. The environment mainly includes a node list, edge list, and label set. Among them, the node list records the host index, and the label

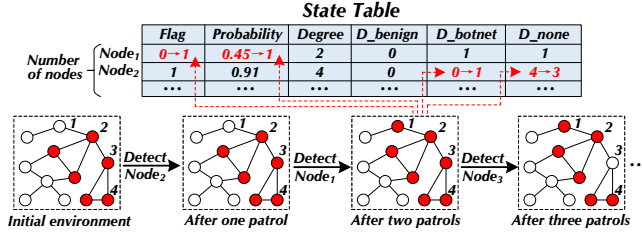


Figure 2: The state table design. The red nodes represent bots.

Table 1: The classification results of the SOTA model, and our calibration results (with 20K retrievals) for the Chord botnet. “Base” indicates that there is no benign P2P user.

Metrics	Acc	FPR	FNR	Rec	Pre	F1	AUC	Our Acc
Base	99.87	0.03	1.41	98.59	99.58	99.08	99.98	-
Bamboo	92.23	7.00	18.11	81.89	46.62	59.42	92.76	99.46▲7.22
Broose	92.31	6.99	17.09	82.91	46.96	59.95	92.81	99.54▲7.24
Chord	92.55	7.02	13.23	86.77	47.96	61.77	93.03	99.63▲7.08
Gia	93.01	7.02	6.54	93.46	49.83	65.00	93.46	99.79▲6.78
Kademlia	92.35	7.02	16.06	83.94	47.15	60.38	92.91	99.48▲7.13
Koorde	92.81	7.01	9.53	90.47	49.02	63.58	93.30	99.74▲6.93
Nice	92.55	7.02	13.23	86.77	47.96	61.76	93.03	99.65▲7.10

set stores the corresponding ground truth (“benign” as “0” and “botnet” as “1”). Figure 2 shows the input structure, it consists of 6-dimension properties, *i.e.*, “Flag”: records whether the node has been reviewed; “Probability”: refers to the confidence (initialized by the GNN model and will be updated with the detection result); “Degree”: indicates the degree of this node (undirected); “D_benign”, “D_botnet”, and “D_none” represent the number of viewed benign, botnet neighbors, and unreviewed neighbors of this node.

Agent and Action. To decide on the next node to check, we design a tailor-made agent architecture, which cascades a series of Linear layers and ReLU activations. Subsequently, the output vector (V_o) will subtract the product of the maximum (*e.g.*, float.max) and flag vector V_f (from input). Also, this process can suppress the probability of selecting the nodes that have already been detected. Finally, that difference vector is cascaded with the softmax layer and sampled by “Categorical distribution”, *i.e.*, $\text{Action} = \text{Categorical}(\text{Softmax}(V_o - V_f))$, to get the next detected node.

Reward and Training. When the agent makes a decision, it will detect whether there is malicious behavior in the node traffic and update the environment and state tables. Meanwhile, the reward value will be calculated based on node correction and boundary (between benign and botnet) discovery. Subsequently, we leverage the Policy Gradient method [27] to optimize the agent.

3 EXPERIMENTS

Datasets. The datasets include botnet communication graphs from Zhou et al. [36] and 7 common P2P traffic generated based on OverSim [2]. The former produces four P2P-based botnets topologies (*i.e.*, Koorde [14], Kademlia [18], Chord [24], Leet-chord [13]) and collects two real-world botnets (*i.e.*, P2P and C2 [6]). The latter includes 7 common P2P protocols, *i.e.*, Bamboo, Broose, Chord, Gia, Kademlia, Koorde, and Nice. Consistent with previous works [19, 36], we synthesize communication graphs including botnet and real-world background traffic, with the generated legitimate P2P traffic.

Compare with SOTA. In Table 1 for Chord botnet, the performance of GCN [36] significantly drops by comparing 7 types of

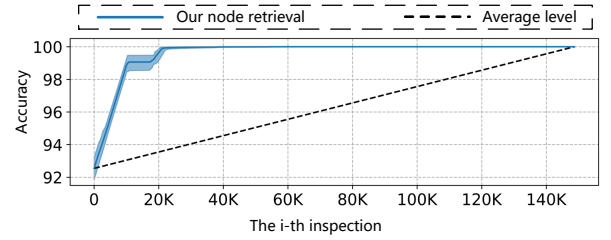


Figure 3: The accuracy curve with the calibration process.

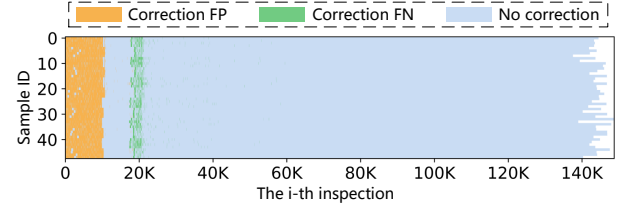


Figure 4: The calibration process visualization.

benign P2P with the “Base” (no benign P2P embedded). For instance, the FPR directly dropped 7 percent, noteworthy, given that the embedded legitimate P2P nodes are around 10K (occupying 7.14% of 140K nodes), 7% FPR means >98% benign P2P users will be misidentified. For the last column, we report the calibration accuracy with 20K retrievals. It is clear that our model basically improves the accuracy to 99.9%~100% by only retrieving about $\frac{1}{7}$ nodes (*i.e.*, 20K/140K). We draw the accuracy growth curve (denote blue line) in Figure 3, it is clear that the proposal can achieve superior performance compared to random sampling (black line).

Calibration Process. We visualize the RL-based calibration process in Figure 4, where the yellow (green) mark refers to correcting a false positive (false negative), and the blue mark denotes correct predictions. As expected, most false positives and false negatives are corrected at an early stage, indicating the RL agent could quickly locate those nodes that may be misidentified.

Deep Insights. We use the Integrated Gradients method [26] from Captum [4] to generate attribution matrices for interpretability. We present the details of the attribution map and the calibrated node instances in the online repository¹. We find that the retrieval process first depends on “Degree”, then “D_botnet” dominates at ~20K, and subsequently mainly related to “Probability”. For three instances with respect to the 1-th, 4K-th, and 20K-th node calibration. We observe that the RL agent prefers nodes with a larger degree in the early stage, given such nodes could bring more rewards from boundary discovery. At about 20K-th retrievals, the RL model tends to select those nodes whose neighbors are mainly botnets.

4 CONCLUSION

In this paper, we reveal the limitations of the SOTA graph-based botnet detection when P2P botnets and legitimate P2P users exist simultaneously. To this end, we propose an RL-based approach to effectively perform node retrieval to correct botnet recognition. We conduct extensive evaluations, the results show that our proposal can improve the ACC to 99.9% when only query $< \frac{1}{7}$ nodes. Our research provides an effective solution for goal-specific node retrieval on the large-scale graph.

¹<https://github.com/Secbrain/TNT>

REFERENCES

- [1] Manos Antonakakis, Tim April, Michael Bailey, Matt Bernhard, Elie Bursztein, Jaime Cochran, Zakir Durumeric, J. Alex Halderman, Luca Invernizzi, Michalis Kallitsis, Deepak Kumar, Chaz Lever, Zane Ma, Joshua Mason, Damian Menscher, Chad Seaman, Nick Sullivan, Kurt Thomas, and Yi Zhou. 2017. Understanding the Mirai Botnet. In *USENIX Security Symposium*. USENIX Association, 1093–1110.
- [2] Ingmar Baumgart, Bernhard Heep, and Stephan Krause. 2009. OverSim: A Scalable and Flexible Overlay Framework for Simulation and Real Network Applications. In *Peer-to-Peer Computing*. IEEE, 87–88.
- [3] Juan Caballero, Pongsin Poosankam, Christian Kreibich, and Dawn Xiaodong Song. 2009. Dispatcher: enabling active botnet infiltration using automatic protocol reverse-engineering. In *CCS*. ACM, 621–634.
- [4] Captum. 2022. Captum: Model Interpretability for PyTorch. <https://captum.ai/>.
- [5] Chia Yuan Cho, Domagoj Babic, Eui Chul Richard Shin, and Dawn Song. 2010. Inference and analysis of formal models of botnet command and control protocols. In *CCS*. ACM, 426–439.
- [6] Sebastián García, Martin Grill, Jan Stiborek, and Alejandro Zunino. 2014. An empirical comparison of botnet detection methods. *Comput. Secur.* 45 (2014), 100–123.
- [7] Steven Gianvecchio, Mengjun Xie, Zhengyu Wu, and Haining Wang. 2008. Measurement and Classification of Humans and Bots in Internet Chat. In *USENIX Security Symposium*. USENIX Association, 155–170.
- [8] Guofoi Gu, Roberto Perdisci, Junjie Zhang, and Wenke Lee. 2008. BotMiner: Clustering Analysis of Network Traffic for Protocol- and Structure-Independent Botnet Detection. In *USENIX Security Symposium*. USENIX Association, 139–154.
- [9] Guofoi Gu, Phillip A. Porras, Vinod Yegneswaran, Martin W. Fong, and Wenke Lee. 2007. BotHunter: Detecting Malware Infection Through IDS-Driven Dialog Correlation. In *USENIX Security Symposium*. USENIX Association.
- [10] Guofoi Gu, Junjie Zhang, and Wenke Lee. 2008. BotSniffer: Detecting Botnet Command and Control Channels in Network Traffic. In *NDSS*. The Internet Society.
- [11] Stephen Herwig, Katura Harvey, George Hughey, Richard Roberts, and Dave Levin. 2019. Measurement and Analysis of Hajime, a Peer-to-peer IoT Botnet. In *NDSS*. The Internet Society.
- [12] Grégoire Jacob, Ralf Hund, Christopher Kruegel, and Thorsten Holz. 2011. JACK-STRAWS: Picking Command and Control Connections from Bot Traffic. In *USENIX Security Symposium*. USENIX Association.
- [13] Márk Jelasity and Vilmos Bilicki. 2009. Towards Automated Detection of Peer-to-Peer Botnets: On the Limits of Local Approaches. In *LEET*. USENIX Association.
- [14] M. Frans Kaashoek and David R. Karger. 2003. Koorde: A Simple Degree-Optimal Distributed Hash Table. In *IPTPS (Lecture Notes in Computer Science, Vol. 2735)*. Springer, 98–107.
- [15] Thomas N. Kipf and Max Welling. 2017. Semi-Supervised Classification with Graph Convolutional Networks. In *ICLR (Poster)*. OpenReview.net.
- [16] Zhaoxuan Li et al. 2024. metaNet: Interpretable unknown mobile malware identification with a novel meta-features mining algorithm. *Comput. Networks* (2024), 110563.
- [17] Xi Ling et al. 2024. DDoSMiner: An Automated Framework for DDoS Attack Characterization and Vulnerability Mining. In *ACNS (2) (Lecture Notes in Computer Science, Vol. 14584)*. Springer, 283–309.
- [18] Petar Maymounkov and David Mazières. 2002. Kademlia: A Peer-to-Peer Information System Based on the XOR Metric. In *IPTPS (Lecture Notes in Computer Science, Vol. 2429)*. Springer, 53–65.
- [19] Shishir Nagaraja, Prateek Mittal, Chi-Yao Hong, Matthew Caesar, and Nikita Borisov. 2010. BotGrep: Finding P2P Bots with Structured Graph Analysis. In *USENIX Security Symposium*. USENIX Association, 95–110.
- [20] Andreas Pitsillidis, Kirill Levchenko, Christian Kreibich, Chris Kanich, Geoffrey M. Voelker, Vern Paxson, Nicholas Weaver, and Stefan Savage. 2010. Botnet Judo: Fighting Spam with Itself. In *NDSS*. The Internet Society.
- [21] Tohid Shekari, Celine Irvine, Alvaro A. Cárdenas, and Raheem Beyah. 2021. MaMIoT: Manipulation of Energy Market Leveraging High Wattage IoT Botnets. In *CCS*. ACM, 1338–1356.
- [22] Saleh Soltan, Prateek Mittal, and H. Vincent Poor. 2018. BlackIoT: IoT Botnet of High Wattage Devices Can Disrupt the Power Grid. In *USENIX Security Symposium*. USENIX Association, 15–32.
- [23] Zhuoxue Song et al. 2023. I2RNN: An Incremental and Interpretable Recurrent Neural Network for Encrypted Traffic Classification. *IEEE Transactions on Dependable and Secure Computing* (2023).
- [24] Ion Stoica, Robert Tappan Morris, David R. Karger, M. Frans Kaashoek, and Hari Balakrishnan. 2001. Chord: A scalable peer-to-peer lookup service for internet applications. In *SIGCOMM*. ACM, 149–160.
- [25] Gianluca Stringhini, Thorsten Holz, Brett Stone-Gross, Christopher Kruegel, and Giovanni Vigna. 2011. BOTMAGNIFIER: Locating Spambots on the Internet. In *USENIX Security Symposium*. USENIX Association.
- [26] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. 2017. Axiomatic Attribution for Deep Networks. In *ICML (Proceedings of Machine Learning Research, Vol. 70)*. PMLR, 3319–3328.
- [27] Richard S. Sutton, David A. McAllester, Satinder Singh, and Yishay Mansour. 1999. Policy Gradient Methods for Reinforcement Learning with Function Approximation. In *NIPS*. The MIT Press, 1057–1063.
- [28] Zhen Zeng, Chun-Jen Chung, and Liguang Xie. 2022. Security Challenges for Modern Data Centers with IoT: A Preliminary Study. In *WWW (Companion Volume)*. ACM, 555–562.
- [29] Menghao Zhang, Guanyu Li, Shicheng Wang, Chang Liu, Ang Chen, Hongxin Hu, Guofoi Gu, Qi Li, Mingwei Xu, and Jianping Wu. 2020. Poseidon: Mitigating Volumetric DDoS Attacks with Programmable Switches. In *NDSS*. The Internet Society.
- [30] Ziming Zhao et al. 2024. CMD: Co-Analyzed IoT Malware Detection and Forensics via Network and Hardware Domains. *IEEE Trans. Mob. Comput.* 23, 5 (2024), 5589–5603.
- [31] Ziming Zhao et al. 2024. DDoS family: A novel perspective for massive types of DDoS attacks. *Comput. Secur.* 138 (2024), 103663.
- [32] Ziming Zhao et al. 2024. Trident: A Universal Framework for Fine-Grained and Class-Incremental Unknown Traffic Detection. In *WWW*. ACM, 1608–1619.
- [33] Ziming Zhao, Zhaoxuan Li, et al. 2023. ERNN: Error-Resilient RNN for Encrypted Traffic Detection towards Network-Induced Phenomena. *IEEE Transactions on Dependable and Secure Computing* (2023).
- [34] Ziming Zhao, Zhaoxuan Li, Zhuoxue Song, and Fan Zhang. 2023. Work-in-Progress: Towards Real-Time IDS via RNN and Programmable Switches Co-Designed Approach. In *RTSS*. IEEE.
- [35] Ziming Zhao, Zhuotao Liu, Huan Chen, Fan Zhang, Zhuoxue Song, and Zhaoxuan Li. 2024. Effective DDoS Mitigation via ML-Driven In-network Traffic Shaping. *IEEE Transactions on Dependable and Secure Computing* (2024).
- [36] Jiawei Zhou, Zhiying Xu, Alexander M. Rush, and Minlan Yu. 2020. Automating Botnet Detection with Graph Neural Networks. *CoRR* abs/2003.06344 (2020).